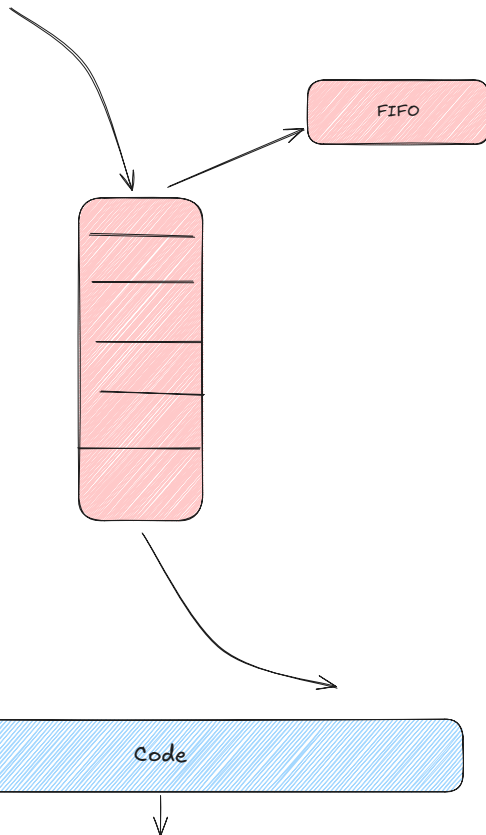


Queue



```
class Queue:
    def __init__(self):
        self.items = []

    def is_empty(self):
        return len(self.items) == 0

    def enqueue(self, item):
        self.items.append(item)

    def dequeue(self):
        if self.is_empty():
            raise IndexError("Queue is empty")
        x = self.items.pop(0)
        return x

    def front(self):
        if self.is_empty():
            raise IndexError("Queue is empty")
        return self.items[0]

    def rear(self):
        if self.is_empty():
            raise IndexError("Queue is empty")
        return self.items[-1]

    def size(self):
        return len(self.items)

queue = Queue()
queue.enqueue(1)
queue.enqueue(2)
queue.enqueue(3)
queue.enqueue(4)

print(f"Dequeued Element: {queue.dequeue()}")
print(f"Front Element: {queue.front()}")
print(f"Queue Size: {queue.size()}")
print(f"Rear Element: {queue.rear()}")
print(f"Is Queue Empty?: {queue.is_empty()}")
```

Enqueue(x)
Dequeue()
Front()
Rear()
Is_empty()
Size()

Frame

Queue Operations

1. **Enqueue(x)**

- * Inserts an element 'x' at the **rear (end)** of the queue.
- * Follows FIFO order.
- * Example:

```
'''text
Queue: [10, 20]
Enqueue(30)
Queue: [10, 20, 30]
'''
```

2. **Dequeue()**

- * Removes and returns the element from the **front** of the queue.
- * The first inserted element is removed first.
- * Example:

```
'''text
Queue: [10, 20, 30]
Dequeue()
Removed: 10
Queue: [20, 30]
'''
```

3. **Front()**

- * Returns the element at the **front** of the queue without removing it.
- * Example:

```
'''text
Queue: [10, 20, 30]
Front() → 10
'''
```

4. **Rear()**

- * Returns the element at the **rear (last position)** of the queue without removing it.
- * Example:

```
'''text
Queue: [10, 20, 30]
Rear() → 30
'''
```

5. **Is_Empty()**

- * Checks whether the queue contains any elements.
- * Returns 'True' if the queue is empty, otherwise 'False'.
- * Example:

```
'''text
Queue: []
Is_Empty() → True
'''
```

6. **Size()**

- * Returns the total number of elements currently present in the queue.
- * Example:

```
'''text
Queue: [10, 20, 30]
Size() → 3
'''
```

Queue Definition

- * A **Queue** is a linear data structure that follows the **FIFO (First In First Out)** principle.
- * The element inserted first is removed first.
- * Insertion is performed at the **rear**, and deletion is performed at the **front**.

FIFO Example

```
'''text
Enqueue(10)
Enqueue(20)
Enqueue(30)
```

Queue: [10, 20, 30]

```
Dequeue() → 10
Dequeue() → 20
Dequeue() → 30
'''
```